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ASE LAB 8

Step 1:

- Firstly Open the docker desktop
- Then run this command:

```
docker run -d --name sonarqube -e
SONAR_ES_BOOTSTRAP_CHECKS_DISABLE=true -p 9000:9000
sonarqube:latest
```

- Wait until the command and then run: `docker ps`

If u see output like this:

```
C:\Users\Dell>docker run -d --name sonarqube -e SONAR_ES_BOOTSTRAP_CHECKS_DISABLE=true -p 9000:9000 sonarqube:latest
Unable to find image 'sonarqube:latest' locally
latest: Pulling from library/sonarqube
4f4fb700ef54: Pull complete
3edbd929c593: Pull complete
e77c97483e9c: Pull complete
e0cb59978564: Pull complete
89cc57adce47: Pull complete
7130478a3071: Pull complete
edd1ed89f0d4: Pull complete
fd0be37e3cf8: Download complete
ae7e64a49f75: Download complete
Digest: sha256:4905574ab8584dcac1a06c01cdc43a4ce9e369723ac4a7ab1d029c1bf71e6fe9
Status: Downloaded newer image for sonarqube:latest
a2566430c8e36ca39173108d125b209c9bfd710964e488986a3da0595b9e1950

C:\Users\Dell>docker ps
```

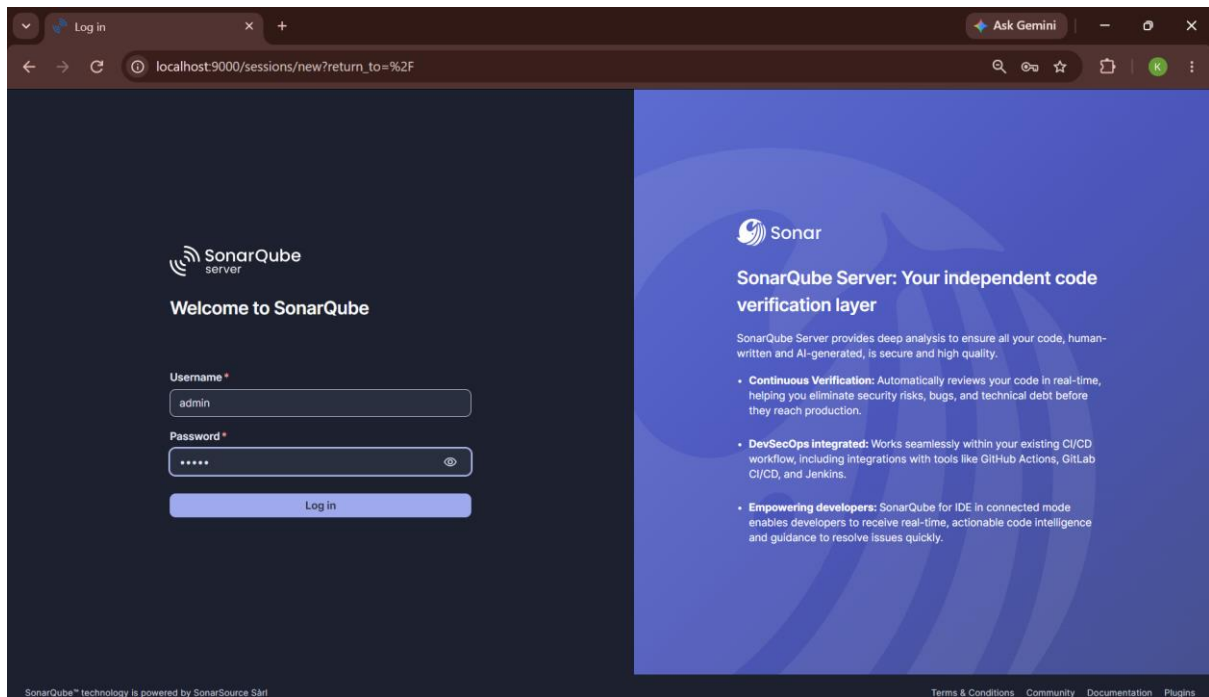
CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
a2566430c8e3	sonarqube:latest	"/opt/sonarqube/dock..."	14 seconds ago	Up 13 seconds	0.0.0.0:9000->9000/tcp
8f9284531f83	gcr.io/k8s-minikube/kicbase:v0.0.50	"/usr/local/bin/entr..."	6 weeks ago	Up 58 minutes	127.0.0.1:52146->22/tcp
p, 127.0.0.1:52142->2376/tcp, 127.0.0.1:52144->5000/tcp, 127.0.0.1:52145->8443/tcp, 127.0.0.1:52143->32443/tcp					sonarqube minikube

```
C:\Users\Dell>
```

Then The image has been installed successfully

Step 2: Open sonarqube using localhost

<http://localhost:9000>



➔ Then login using the credentials:

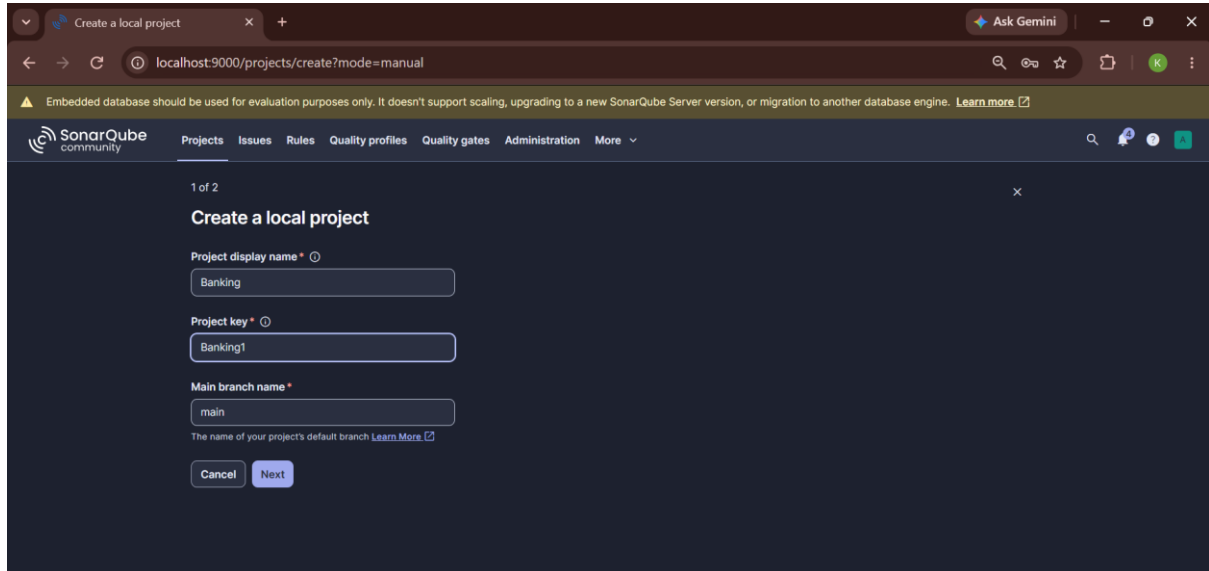
Username: admin

Password: admin

Step 3: Create the SonarQube project

Project Name: Banking

Project Key: Banking



STEP 4 — Create your Java project folder

The structure should be like this:

ASE 8

```
|
|
|— sonar-project.properties
|
|
|— src
|
|
|— Main.java
```

sonar-project.properties:

sonar.projectKey=Banking

sonar.projectName=Banking

sonar.sources=src

sonar.host.url=http://localhost:9000

Main.java: (can write your own code if not just copy paste this):

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        int a = 10;
```

```
        int b = 20;
```

```
        System.out.println("Result: " + calculate(a, b));
```

```
        String name = null;
```

```
        if (name != null) {
```

```
            System.out.println(name);
```

```
        }
```

```
        int unused = 100;
```

```
        for (int i = 0; i < 10; i++) {
```

```
            System.out.println(i);
```

```
        }
```

```
    }
```

```
    public static int calculate(int a, int b) {
```

```
        return a + b;
```

```
}
```

```
public static void badMethod() {
```

```
    int x = 10;
```

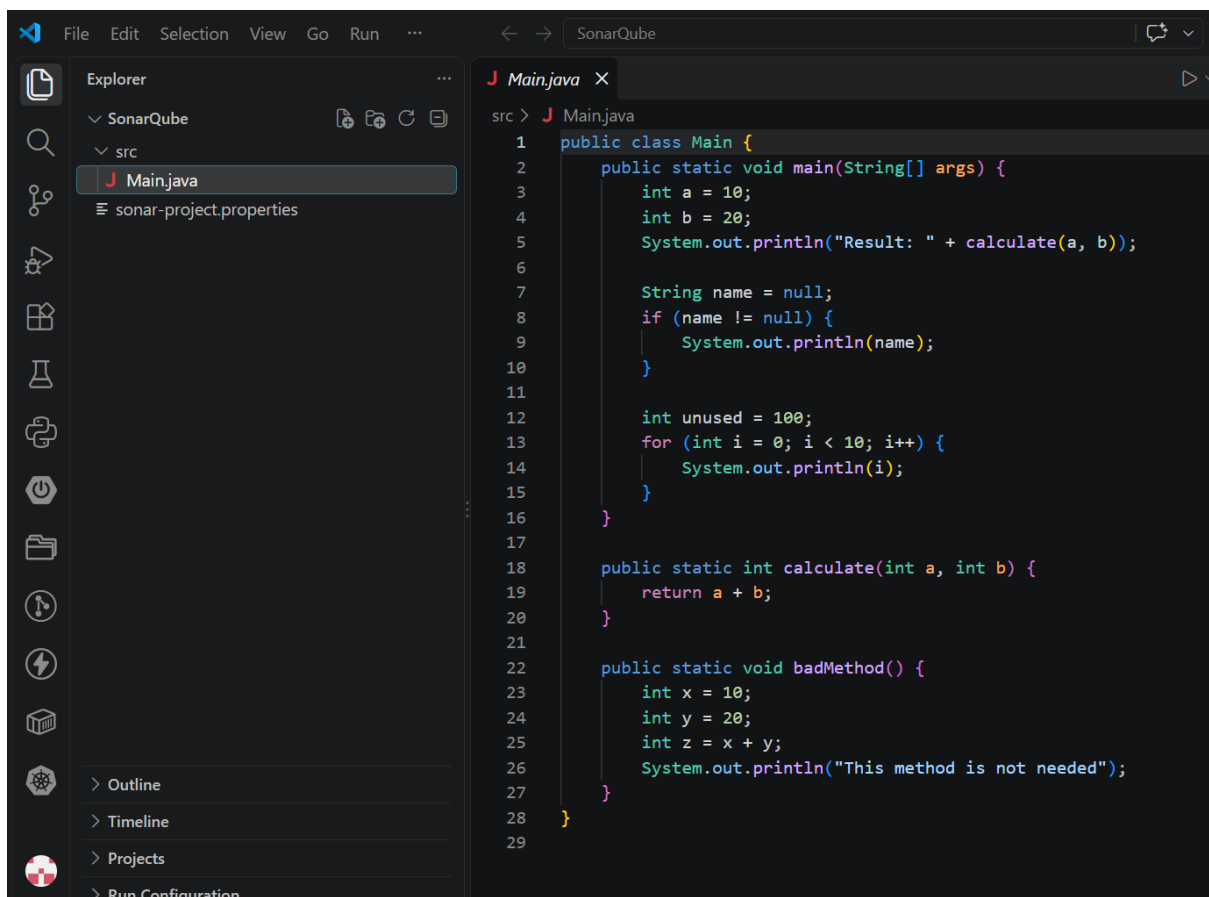
```
    int y = 20;
```

```
    int z = x + y;
```

```
    System.out.println("This method is not needed");
```

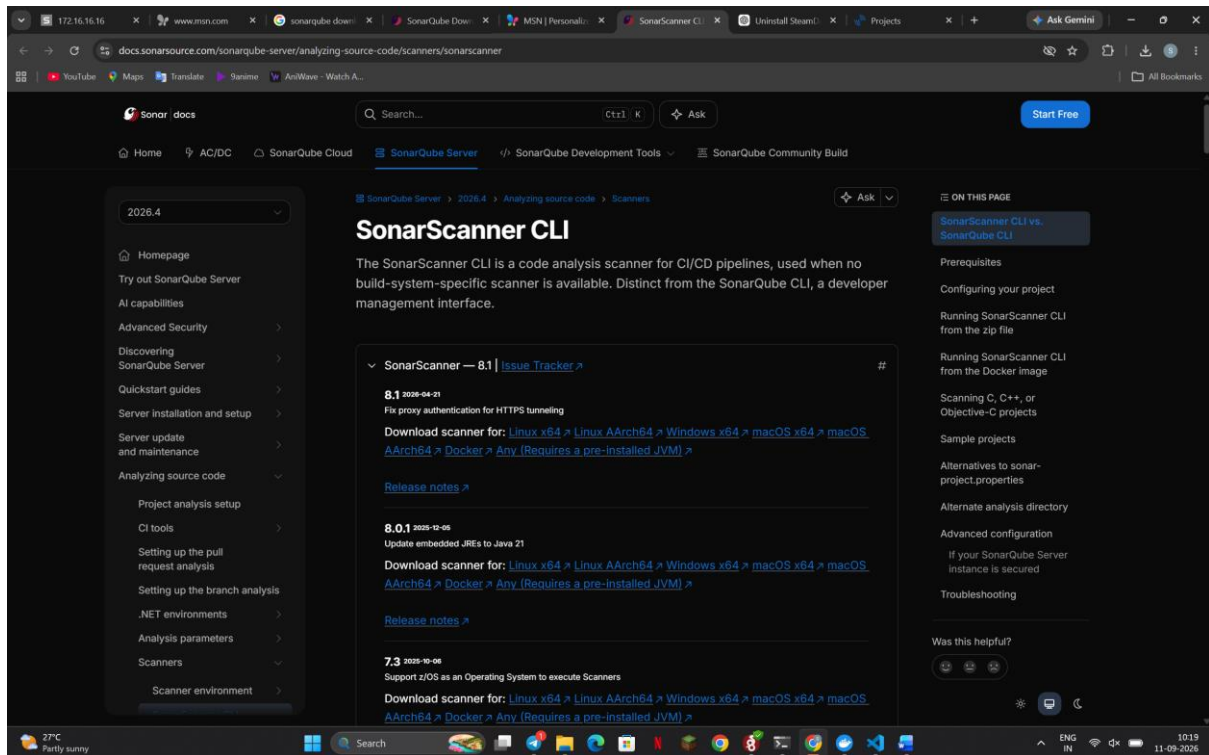
```
}
```

```
}
```



Step 5: Install SonarScanner

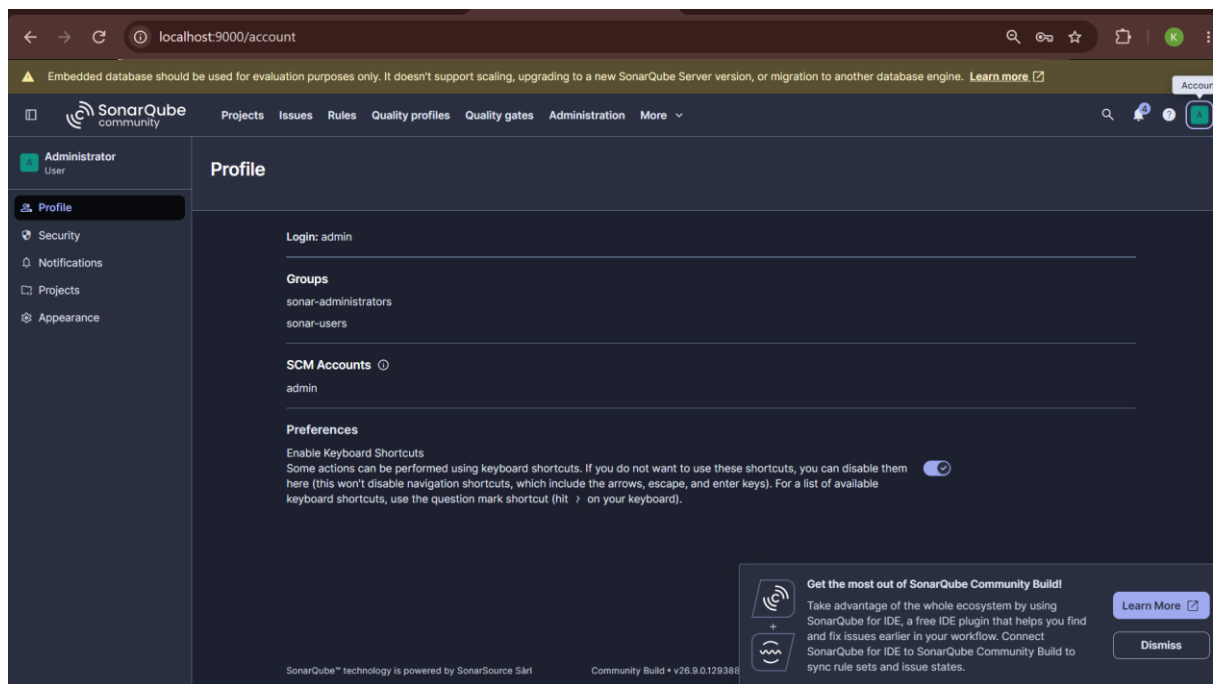
<https://docs.sonarsource.com/sonarqube-server/analyzing-source-code/scanners/sonarscanner>



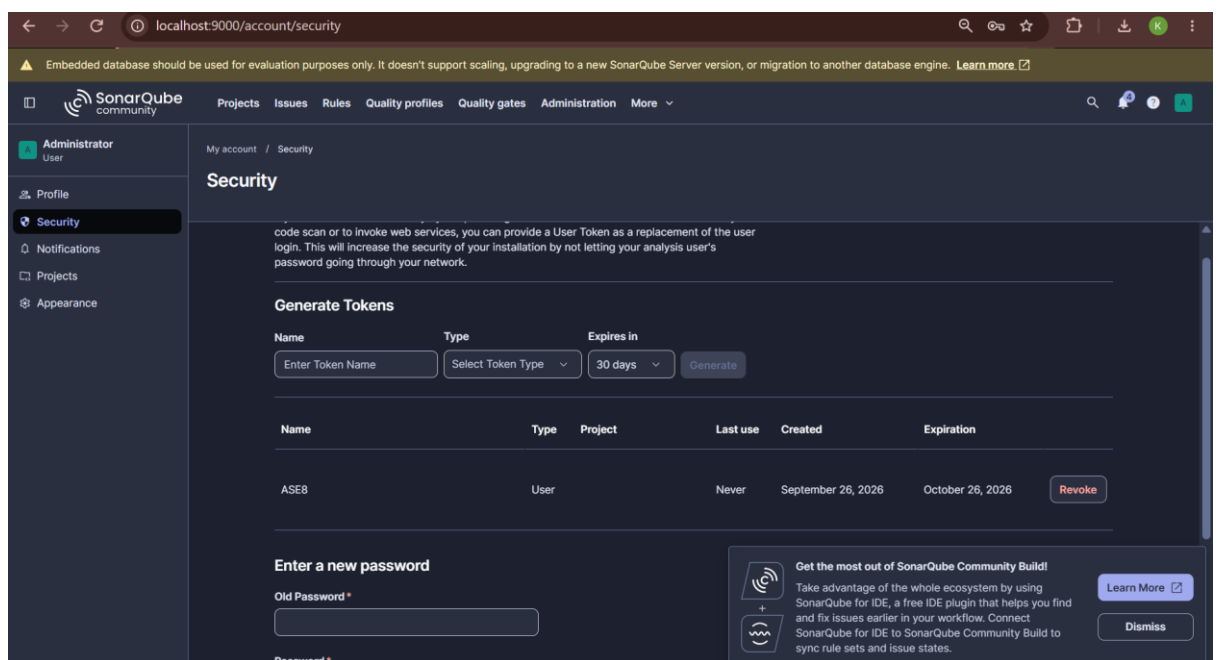
➔ To check if its installed correctly use: `sonar-scanner --version`

Step 6: Generate SonarQube authentication token

- ➔ Go to the localhost:9000, login using the credentials
- ➔ After logging in open your profile on the right corner



➔ Then go to the Security option in the left



➔ Give a name to the token and then select the type as User token and generate it

➔ Copy the token and open the cmd and run
set SONAR_TOKEN=YOUR_TOKEN

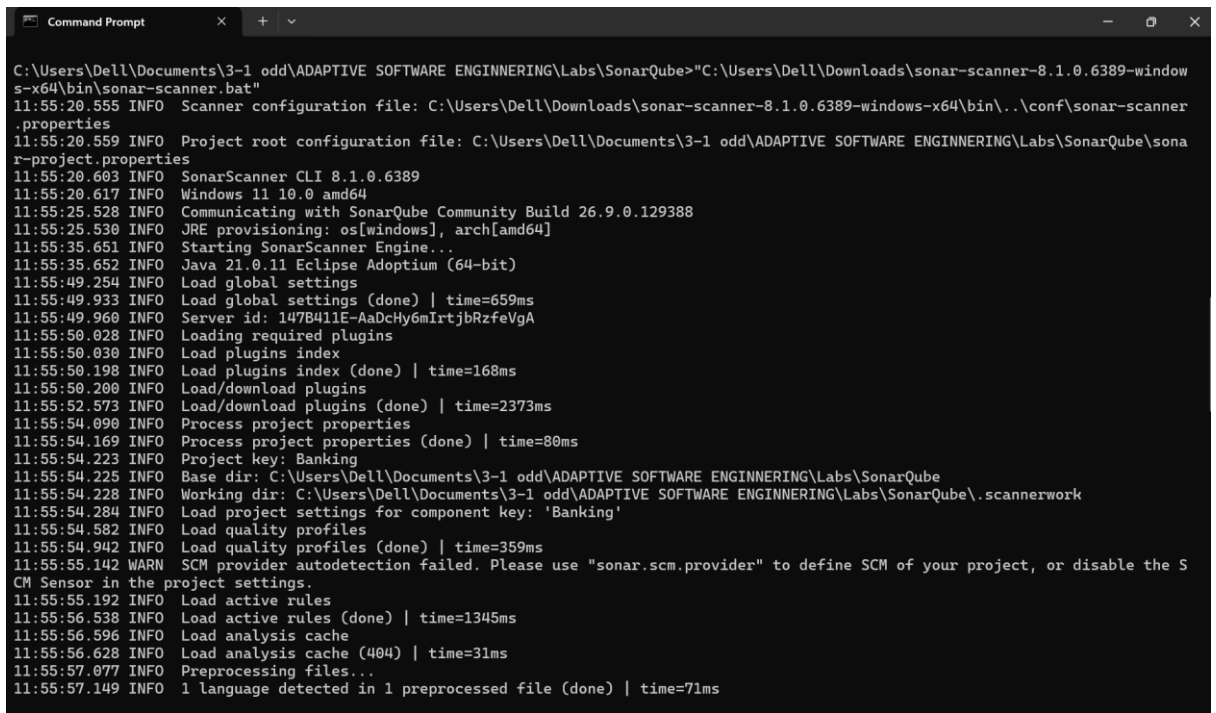
Step 7: Go to your project directory

➔ Open cmd and then go to your project directory ASE 8

Step 8: Run SonarScanner

➔ After going to your directory in cmd run the command:

sonar-scanner



```
C:\Users\Dell\Documents\3-1 odd\ADAPTIVE SOFTWARE ENGINEERING\Labs\SonarQube>"C:\Users\Dell\Downloads\sonar-scanner-8.1.0.6389-windows-x64\bin\sonar-scanner.bat"
11:55:20.555 INFO Scanner configuration file: C:\Users\Dell\Downloads\sonar-scanner-8.1.0.6389-windows-x64\bin\..\conf\sonar-scanner.properties
11:55:20.559 INFO Project root configuration file: C:\Users\Dell\Documents\3-1 odd\ADAPTIVE SOFTWARE ENGINEERING\Labs\SonarQube\sonar-project.properties
11:55:20.603 INFO SonarScanner CLI 8.1.0.6389
11:55:20.617 INFO Windows 11 10.0 amd64
11:55:25.528 INFO Communicating with SonarQube Community Build 26.9.0.129388
11:55:25.530 INFO JRE provisioning: os[windows], arch[amd64]
11:55:35.651 INFO Starting SonarScanner Engine...
11:55:35.652 INFO Java 21.0.11 Eclipse Adoptium (64-bit)
11:55:49.254 INFO Load global settings
11:55:49.933 INFO Load global settings (done) | time=659ms
11:55:49.960 INFO Server id: 147B411E-AaDcHy6mIrtjbRzfeVgA
11:55:50.028 INFO Loading required plugins
11:55:50.030 INFO Load plugins index
11:55:50.198 INFO Load plugins index (done) | time=168ms
11:55:50.200 INFO Load/download plugins
11:55:52.573 INFO Load/download plugins (done) | time=2373ms
11:55:54.090 INFO Process project properties
11:55:54.169 INFO Process project properties (done) | time=80ms
11:55:54.223 INFO Project key: Banking
11:55:54.225 INFO Base dir: C:\Users\Dell\Documents\3-1 odd\ADAPTIVE SOFTWARE ENGINEERING\Labs\SonarQube
11:55:54.228 INFO Working dir: C:\Users\Dell\Documents\3-1 odd\ADAPTIVE SOFTWARE ENGINEERING\Labs\SonarQube\scannerwork
11:55:54.284 INFO Load project settings for component key: 'Banking'
11:55:54.582 INFO Load quality profiles
11:55:54.942 INFO Load quality profiles (done) | time=359ms
11:55:55.142 WARN SCM provider autodetection failed. Please use "sonar.scm.provider" to define SCM of your project, or disable the SCM Sensor in the project settings.
11:55:55.192 INFO Load active rules
11:55:56.538 INFO Load active rules (done) | time=1345ms
11:55:56.596 INFO Load analysis cache
11:55:56.628 INFO Load analysis cache (404) | time=31ms
11:55:57.077 INFO Preprocessing files...
11:55:57.149 INFO 1 language detected in 1 preprocessed file (done) | time=71ms
```

```
Command Prompt
11:56:08.572 INFO Start fetching files for the text and secrets analysis
11:56:09.448 INFO Using Git CLI to retrieve dirty files
11:56:09.584 WARN Retrieving only language associated files, make sure to run the analysis inside a git repository to make use of in
clusions specified via "sonar.text.inclusions"
11:56:09.586 INFO Starting the text and secrets analysis
11:56:09.592 INFO 1 source file to be analyzed for the text and secrets analysis
11:56:09.608 INFO 1/1 source file has been analyzed for the text and secrets analysis
11:56:09.610 INFO Sensor TextAndSecretsSensor [text] (done) | time=1828ms
11:56:09.614 INFO ----- Run sensors on project
11:56:09.903 INFO Sensor JaCoCo Aggregate XML Report Importer [jacoco]
11:56:09.904 INFO Sensor JaCoCo Aggregate XML Report Importer [jacoco] (done) | time=0ms
11:56:09.904 INFO Sensor IaC Project Sensor [iac]
11:56:09.906 INFO Sensor IaC Project Sensor [iac] (done) | time=3ms
11:56:09.907 INFO Sensor JavaProjectSensor [java]
11:56:09.913 INFO Sensor JavaProjectSensor [java] (done) | time=7ms
11:56:09.914 INFO Sensor Zero Coverage Sensor
11:56:09.924 INFO Sensor Zero Coverage Sensor (done) | time=10ms
11:56:09.926 INFO Sensor Java CPD Block Indexer
11:56:09.946 INFO Sensor Java CPD Block Indexer (done) | time=21ms
11:56:09.946 INFO ----- Gather SCA dependencies on project
11:56:09.954 INFO Dependency analysis skipped
11:56:09.959 INFO SCM Publisher No SCM system was detected. You can use the 'sonar.scm.provider' property to explicitly specify it.
11:56:09.967 INFO CPD Executor Calculating CPD for 1 file
11:56:09.976 INFO CPD Executor CPD calculation finished (done) | time=9ms
11:56:10.353 INFO Analysis report generated in 364ms, dir size=293.4 kB
11:56:10.385 INFO Analysis report compressed in 32ms, zip size=32.4 kB
11:56:11.374 INFO Analysis report uploaded in 989ms
11:56:11.378 INFO ANALYSIS SUCCESSFUL, you can find the results at: http://localhost:9000/dashboard?id=Banking
11:56:11.379 INFO Note that you will be able to access the updated dashboard once the server has processed the submitted analysis re
port
11:56:11.380 INFO More about the report processing at http://localhost:9000/api/ce/task?id=c25cfced-b539-4655-816d-61ca6fdb995b
11:56:11.409 INFO Analysis total time: 18.518 s
11:56:11.461 INFO SonarScanner Engine completed successfully
11:56:11.541 INFO EXECUTION SUCCESS
11:56:11.542 INFO Total time: 50.994s

C:\Users\Dell\Documents\3-1 odd\ADAPTIVE SOFTWARE ENGINEERING\Labs\SonarQube>
```

➔ When you see the execution as successful go the localhost:9000

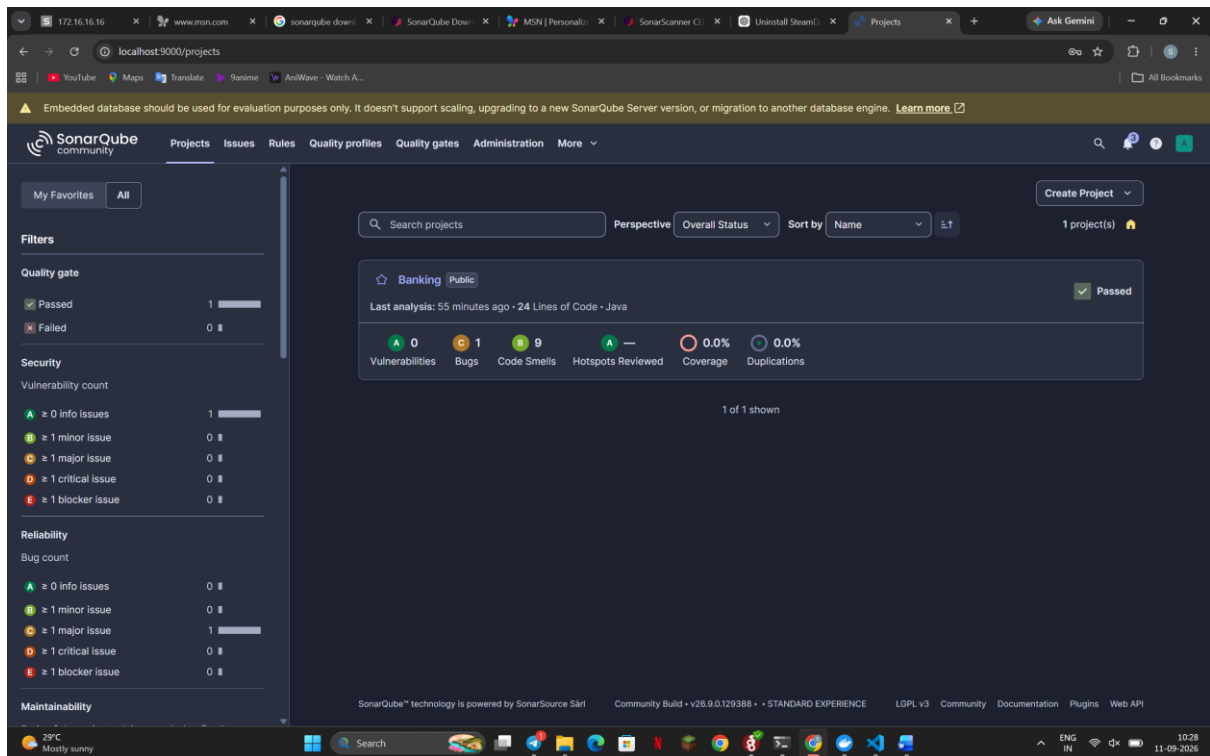
Step 9: Open your project dashboard

Go to:

<http://localhost:9000>

Then:

Projects → Banking



Here u should see your code if its passed or failed